MR LUYOLO MAHAMBA
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CAREER OBJECTIVE

I am a BSc student in Mathematics and Chemistry at the University of Johannesburg, passionate about applying quantitative reasoning, structured problem-solving, and logical thinking in software development. I have a growing interest in software engineering principles, object-oriented programming, database management, and user interface design.

My programming background is preparing to contribute to designing, developing, and maintaining robust software applications. I aim to grow professionally and personally through entry-level roles, graduate programs, and internship opportunities, while contributing with commitment, curiosity, and teamwork.

EDUCATION

Senior Certificate (Matric)
School: Sithukuthezi Secondary School

Bachelor of Science in Chemistry and Mathematics
University: University of Johannesburg Duration: 2023
-Expected Completion 2026

Key Modules:

- Calculus I & II (Rates of change, optimization, integrals)
- Linear Algebra (Vector spaces, matrices, and systems of equations)
- Differential Equations (Mathematical modelling and forecasting)
- Inorganic & Physical Chemistry (Critical reasoning and system analysis)
- Organic Chemistry
- Introduction to Algorithm Development (C++)
- Introduction to Data Structures (C++)

- Object-Oriented Programming
- Data Communications

PROJECTS

Inventory Management System (IMS)
Developed a JavaFX desktop application to help small businesses manage products, suppliers, and transactions.

- Designed intuitive UI with Scene Builder
- Built core features for managing inventory, transaction logging, and supplier tracking
- Integrated a MySQL database with JDBC, ensuring data integrity and persistence
- Implemented robust error handling and input validation

Tech Stack: Java, JavaFX, JDBC, MySQL, Scene Builder
GitHub: github.com/Luyo200/Inventory-Management-System

SKILLS

Mathematical & Analytical Skills

- Strong foundation in functions, calculus, and systems modelling
- Analytical problem-solving using algebraic and logical reasoning
- Confident interpreting mathematical structures and trends in data
- Solid grasp of physics concepts to support quantitative analysis

Programming & Technology

- **Languages**: Java, C++, JavaScript, HTML, CSS

- **Frameworks**: JavaFX
- **Tools**: GitHub, Eclipse, VS Code, Scene Builder, Notepad++
- **Microsoft Office**: Word, Excel, PowerPoint

Soft Skills

- Detail-oriented and reliable, thrives in teams and excels independently.
- Eager to learn and adapt quickly in dynamic environments

Languages

- Fluent in English, IsiZulu, and IsiXhosa

INTERESTS

- Designing and developing applications across platforms, including mobile apps, web applications, and desktop software
- Implementing mathematical models and algorithms to enhance app performance and functionality
- Creating simulations, automation tools, and utilities to support seamless user experiences
- Leveraging digital transformation and process automation to streamline development workflows
- Applying logic-based problem-solving to build scalable, user-friendly, and robust software solutions